

Andrew Cheng

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SUMMARY

Second-Year Electrical Computer Engineering Student at Santa Clara University with over 5 years of experience in building computers and understanding performance metrics with components, A self taught learner driven by exploration in analog systems, digital systems. Looking for roles in Analog design, Digital design, IT support.

EDUCATION

Santa Clara University — Santa Clara, CA Expected June 2029

Bachelor of Science, Electrical and Computer Engineering

Major GPA: 3.60 | Overall GPA: 3.39

Relevant Coursework: Intro to Programming, Intro to Engineering, Intro to Digital Logic, Emerging areas of Electrical Engineering, Advanced Programming, Pre-calculus, Calculus I

SKILLS

Languages: Python, C, C++, HTML, PHP, VHDL

Operating Systems: Windows, macOS, Linux, Unix

Software & Tools: Arduino IDE, KiCad, MATLAB, Logisim-Evolution, LT-Spice, Onshape, Quartus, Text-editor, TinkerCAD, TrueNAS, VS Code, Vim, Terminal, Canva, Google Docs, Google Slides

Hardware: Soldering, Hardware Prototyping, PCB design, Schematic design, FPGA Design

PROJECTS

360° Rotating Christmas Tree

Team project: a motorized holiday decoration built as a gift.

- Designed and built a spinning Christmas tree using an Arduino Uno programmed in C++, integrating photoresistors, LEDs, and motors and Arduino Uno.
- Drafted circuit schematics to develop the design and created a Gantt chart to keep team members on schedule; sourced build materials to stay within a \$20 budget.
- Delivered a working demo on time and under budget, earning positive feedback from TAs, graduate students, and faculty; disassembled components afterward for reuse in future projects.

Macropad

Custom keypad for quick-access software shortcuts on the go.

- Built a programmable macropad using an Arduino Nano, Cherry MX Black switches, and a 3D-printed chassis.
- Wrote the shortcut firmware in C++, strengthening soldering skills and gaining hands-on microcontroller experience.
- Continuously iterating on key layouts and software commands, with plans for a future CNC-machined metal version.

Network Attached Storage (NAS)

Self-hosted network storage solution to keep data off the cloud.

- Built a TrueNAS server from repurposed e-waste computers, creating a personal network cloud with redundant backups.
- Configured two-way sync between local storage and existing cloud services.
- Built Linux Fundamentals by building data arrays and understanding docker containers and data management.

WORK EXPERIENCE

PC Builder — Self-Employed, San Mateo, CA December 2020 – Present

- Built over 100+ computers in the last 5 years
- Troubleshooting and Problem Solving over 50 hardware/software problems.
- Assessed customer needs and budgets to deliver builds that matched their requirements.

IT Support Intern — Oracle January 2025 – March 2025

- As a Support Intern developed skills in software management in a corporate environment and understanding tech support in big corporate environment with applications of AI, Software deployment to various teams.